MILITARY SPECIFICATION

TEST SET, GUIDED WEAPON AN/DSM-139C

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the AN/DSM-139C Guided Weapon Test Set, herein after referred to as the test set. The test set is for flightline use to perform data link Phase I & II operational readiness check (GO/NO GO) of Walleye weapons and control-monitor sets (pods) defined in 6.1.

2. APPLICABLE DOCUMENTS

2.1 Specifications and Standards. Unless otherwise specified, the following specification and standards of the issue listed in that issue of Department of Defense Index of Specifications and Standards (DoDISS) specified in-the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	-Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	-Preservation-Packaging, Methods of
MIL-E-5400	-Electronic Equipment; Aerospace Guided Missles Applicable Documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-E-17555 -Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of

MIL-T-28800 -Test Equipment for use with Electrical and Electronic Equipment, General Specification for

STANDARDS

FEDERAL

FED-STD-123 -Marking for Shipment (Civil Agencies)

MILITARY

MIL-STD-108 -Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-129 -Marking for Shipment and Storage

MIL-STD-130 -Identification Marking of US Military Property

MIL-STD-454 -Standard General Requirements for Electronic Equipment

MIL-STD-461 -Electromagnetic Emission and Susceptibility Requirements for the control of Electromagnetic Interference

MIL-STD-781 -Reliability Tests: Exponential Distribution

MIL-STD-810 -Environmental Test Methods

MIL-STD-965 -Parts Control Program

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2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

DL 1098AS1OOO	-Drawing List for Guided Weapon Test Set AN/DSM-I39C

PUBLICATIONS

NAVAIR 16-30DSM139C-1	-Maintenance Instructions (Intermediate), Calibration Instructions, Alignment Instructions, Illustrated Parts Breakdown for Guided Weapon Test Set AN/DSM-139C
NAVAIR 16-30DSM139C-IA	-Guided Weapon Test Set AN/DSM-139C and AN/DSM139C (Secret Supplement)

(Copies of specifications, standards, drawings and publications required by contractors In connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Parts and materials. The parts and materials specified in the drawings and related documents for use in the construction of this test set were selected to ensure the fulfillment of all the requirements of this specification.

3.2.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-STD-965, Procedure I.

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3.2.2 Basic conformance. Parts and materials selected for use in the test set shall be as specified in MIL-T-28800. Parts shall be selected from those conforming to MIL-E-5400, unless the parts listed therein will not perform adequately in the intended applications and environments specified herein.

3.2.2.1 Interchangeability. All parts and materials shall meet the interchangeability requirements as specified in MIL-T-28800. The drawings conforming to DL 1098AS1000 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies and parts.

3.3 Design and construction. The test set shall be manufactured in accordance with the drawings and subsidiary specifications conforming to DL 1098AS1000, except that errors of omission or commission on the drawings may be incompatible with the requirements of this specification. In any case, it is a condition of final acceptance that the test set manufactured shall meet all the design, construction, and workmanship requirements of this specification.

3.3.1 Reliability.

3.3.1.1 Operational stability. The test set shall operate as required continuously or intermittently, for not less than 100 hours. During this time, it shall not be necessary to readjust any controls that are inaccessible to the operator during normal use. The test set battery shall be capable of being recharged.

3.3.1.2 Reliability in mean-time-between-failures (MTBF). The test set shall have a specified mean (operating) time between failures of 100 hours when tested in accordance with 4.5.3.

3.3.2 Standard conditions. The following conditions shall be used to establish the normal performance characteristics under standard conditions for performing laboratory bench tests:

a.	Temperature	Room ambient (30 ± 10° C)
b.	Altitude	Normal ground (10,000 ft. maximum)
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power Voltage	Fully charged battery pack (PN 178AS200) or external power source capable of producing 200 mA of current at 10.0 to 15.0 volts dc with a maximum ripple voltage of 1 mV rms.

3.3.2.1 Service conditions. The test set shall meet the storage and operating test requirements of this specification under any environmental service condition or combination of conditions specified in MIL-T-28800 for Class 2 equipment except as modified herein.

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3.3.2.1.1 Storage temperature. Storage temperatures of the test set are limite the battery pack PN 178AS200 to between -40 and +55°C unless the batteries are removed and stored separately.

3.3.2.1.2 Temperature. The test set, with battery pack (PN 178AS200) installed shall meet all operating test requirements of this specification when tested at limited temperatures of -40 to 55°C under the following conditions:

a. The test set, less battery pack (PN 178AS200), shall be stabilized at required test temperature.

b. Battery pack (PN 178AS200) shall be charged at 60 ma for 16 hours at standard conditions.

c. Battery pack (PN 178AS200) shall be placed in test set immediately after charging and the test initiated after the battery has been subjected to the test temperatures for 15 minutes.

3.3.2.1.1.1 Batter Pack. The battery pack shall meet the requirements of Drawing 178AS200.

3.3.2.1.2 Drip-proof. The test set shall meet the test requirements of MIL-STD-108 for drip proof test except the test set shall be inclined only 15 degrees. The test set shall operate during and after exposure.

3.3.2.1.3 Humidity. The test set shall meet the test requirements of MIL-STD-81O, Method 507.2, Procedure IV, except the temperature shall be not greater than 55°C.

3.3.2.1.4 Salt-spray. The test set shall meet the test requirements of MIL-STD-81O, Method 509.2, Procedure I, including Method 509.1 paragraph entitled Summary.

3.3.2.1.5 Explosive atmosphere. The test set shall meet the test requirements of MIL-STD-81O, Method 511.2, Procedure I, except the temperature shall be not greater than 55°C and the altitude shall be not greater than 10,000 feet.

3.3.2.1.6 Vibration. The test set shall meet the test requirements of MIL-T-28800 for Class 2, Style C equipment.

3.3.2.1.7 Shock. The test set shall meet the test requirements of MIL-STD-810, Method 516.3, Procedure I. The test set shall not be operating during the test.

3.3.3 Warm-up time. The electronics assembly shall perform its required functions within the periods specified in 3.3.3.1 through 3.3.3.4 after first turn on:

3.3.3.1 Frequency sweep. The frequency shall be stepped across the F_{OC} ± 2.25 MHz band within 30 seconds after S1 or S2 is depressed. All requirements of the stepped frequency stated in 3.5.1.2.1 shall be met.

3.3.3.2 Detection of RF signal. Lamp, DS-1, shall provide an indication within 5 seconds after appropriate signal is applied and S1 is depressed.

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3.3.3.4 Battery self test. DS-2 shall indicate battery condition within 5 seconds after S1 or S2 is depressed.

3.3.4 Primary input power requirements. The power required shall be one of the following types and shall be not greater than the specified amounts:

a. A fully charged Battery pack which meets the requirements of Drawing 178AS200.

b. An isolated dc power source capable of producing 200 mA of current at 10.0 to 15.0 Vdc, with a maximum ac voltage of 1.0 mV rms.

3.3.5 Electromagnetic interference suppression. The test set shall function properly and survive without degradation or damage to interior component parts when subjected to electromagnetic environments existing on flight decks of aircraft carriers. In addition, electromagnetic radiation and susceptibility requirements of MIL-STD-461, Class A4, shall be met.

3.3.6 Dimensions. The dimensions of the test set case shall be as follows:

a.	Length	-22-1/2 inches
b.	Width	- 19 inches
c.	Depth	- 13-1/2 inches

3.3.7 Weight. The total weight of the test set, including cables, shall be not greater than 55 pounds.

3.3.8 Color and finish. The color and finish of the test set shall be in accordance with DL 1098AS1OOO.

3.3.9 Identification plate and item markings. Nomenclature and identification markings shall be in accordance with MIL-STD-130 and as specified in DL 1098AS1OOO.

3.4 Performance. The performance of the test set shall be as specified herein in addition to the performance requirements specified in DL 1098AS1OOO. Unless otherwise specified herein, these requirements shall apply under both standard (see 3.3.2) and extreme service conditions (see 3.3.2.1), When reduced performance requirements for service conditions are acceptable, such variations are specified herein.

3.4.1 Test set functions. The test set shall perform the functions specified herein. A detailed functional description is provided in NAVAIR 16-30DSM139C-1 and NAVAIR 16-30DSM139C-1A.

3.4.1.1 WPN TEST switch. The WPN TEST switch shall energize all active circuits in the test set to check battery condition, interrogate weapon and receive weapon RF channel signal.

3.4.1.2 Weapon test mode. In the weapon test mode, the test set shall radiate a modulated RF carrier and shall sweep the carrier through the frequency band of the command receivers of the Walleye weapons. Indicator lamp DS1 (green) shall flash on at approximately 1 second intervals, interrupted for 5 to 18 second intervals in each 20 second sweep period.

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3.4.1.3 SELF TEST switch. The SELF TEST switch shall energize circuitry check the condition of the battery pack and all active circuits in the te set in the video detector, the two subcarrier oscillators and the voltage controlled oscillator. A yellow SELF TEST lamp shall flash on at approximately 2 second intervals to indicate the battery pack is in opera condition.

3.4.1.4 POD TEST mode. In the POD TEST mode both WPN and SELF TEST swit shall be depressed simultaneously which shall cause the test set to detec radiated RF from the pod. The green WPN TEST lamp shall light at 1 secon intervals.

3.4.1.5 Phase I/Phase II (TONE/F/H/J/L/N) switch. This switch shall be to select the Phase I labeled as tone) or Phase IX (labeled as F/H/J/L/N

3.4.1.6 Code switches. This switch shall be used in the DPSK mode to set recognition codes.

3.4.1.7 Phase II Mode. In the Phase II mode, the TONE/F/H/J/L/N switch s be in position F, H, J, L, or N, and the code switches used to provide si digit codes to test the weapon and pod.

3.5 Detail requirements.

3.5.1 Guided Weapon Test Set AN/DSM-139C. The test set shall consist of following parts (listed in DL 1098AS1000) and shall be manufactured in accordance with the indicated drawings and all documents listed thereon.

a.	Complete Test Set	1098AS1000
b.	Transit Case	1098AS1200
c.	Electronics Assembly	1098AS1300
d.	Shroud Assembly, Plain	887AS400
e.	Shroud Assembly, EMCON	1098AS500
f.	Shroud Assembly, EMCON	887AS500
g.	Strap Assembly	887AS520
h.	Clamp Assembly	1098AS530
i.	Battery Storage	178AS200

3.5.1.1 Function (electronic assembly). The electronics assembly shall simultaneously radiate a frequency or phase modulated command signal and sh detect a radiated carrier signal which may be unmodulated or frequency modulated with a video signal. The electronics assembly shall perform its radiation and detection functions when no shroud is attached or when any shroud is attached. The plain shroud, when attached to the electronics assembly, shall provide protection from high intensity electromagnetic fi for the detector system in the electronics assembly, and shall absorb RF energy from the antenna on the system under test when the shroud is placed over this antenna. The EMCON shrouds shall perform the same functions as plain shroud except they shall provide additional protection for the dete system and shall have self test capability.

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3.5.1.2 Electronics assembly. The electronics assembly shall contain the radiate and receive antennas, command signal generation subsystem, DPSK subsystem, detector subsystem, 12 volt battery pack, battery test and self test subsystem and the control and indicator subsystem.

3.5.1.2.1 Frequenty (command subsystem). Stepped over a range of F_{oc} -2.25MHz or lower to F_{oc} +2.25MHz or higher. F_{oc} is a frequency which will be specified in the contract (see 6.2.1).

3.5.1.2.1.1 Sweep characteristics. When in the Phase I and Phase II mode, the total sweep period shall be 20 ± 3 seconds.

3.5.1.2.1.2 Modulation. The carrier shall be modulated by 2300 ± 20 Hz, 5400 ± 50 Hz, and 10500 ± 100 Hz sine wave signals in the tone modulation mode. In the DPSK mode the carrier shall be phase modulated in a digital format and include a weapon recognition code.

3.5.1.2.1.3 Carrier deviation. The peak-to-peak carrier deviation shall be 15 to 45 KHz for the 2300 Hz frequency, 25 to 75 KHz for the 5400 Hz frequency, and 50 to 125 KHz for the 10500 Hz frequency in the F_{oc} ± 2.25 MHz band.

3.5.1.2.1.4 Radiated power. Radiated power shall be -40 ± 8 dbm.

3.5.1.2.1.5 Full sweep capability. When in the Phase I and Phase II mode the sweep shall complete its excursion across the F_{oc} ± 2.25 MHz band when S1 and $2 are released while the frequency is within this band, and shall not repeat until either S1 or S2 is again depressed.

3.5.1.2.2 Detector subsystem. When S1 is depressed, the electronics assembly shall detect and indicate the presence of an unmodulated or frequency modulated RF signal as specified in 3.5.1.2.2.1 through 3.5.1.2.2.4.

3.5.1.2.2.1 Frequency (detector subsystem). Any (and all) frequency(ies) in a band of F_{ov} ± 20 MHz shall be detected. F_{ov} is at a frequency which will be specified in the contract.

3.5.1.2.2.2 Power level (detector subsystem). The power level shall be -1 to +10 dBm applied directly to the detector input at P1.

3.5.1.2.2.3 Sensitivity adjustment (detector subsystem). Adjustment of variable resistor, R6, shall provide a threshold sensitivity adjustment over a minimum range of -1 to +3 dBm when the signal is applied directly to the input of the detector at P1.

3.5.1.2.2.4 Detection indication (detector subsystem). An Indication that the electronics assembly has detected a signal, as specified in 3.5.1.2.2 through 3.5.1.2.2.3, shall be provided by indicator lamp, DS-1. The lamp shall start flashing within 5 seconds after application of the RF signal and shall continue with a period of 1 (+1, -.5) second with a duty cycle of 10 to 25 percent.

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3.5.1.2.3 Pod test subsystem. When S1 and S2 are depressed, the electro assembly shall detect and indicate the presence of an unmodulated or freq modulated RF signal as specified in 3.5.1.2.3.1 through 3.5.1.2.3.4.

3.5.1.2.3.1 **Frequency (pod test subsystems).** Any (and all) frequency(ie a band of F_{oc} **±3 MHz shall be detected.** **F_{OC}** is a frequency which will be

specified in the contract.

3.5.1.2.3.2 Power level (pod test subsystem). The power level shall be +5 dbm applied directly to the detector input a P1.

3.5.1.2.3.3 Sensitivity adjustment (pod test subsystem). When R6 is adj to provide threshold sensitivity at +1 dbm for the F_{oc} frequency, R36 sha

provide a sensitivity adjustment range of at least O to -10 dbm for the frequency.

3.5.1.2.3.4 Detection indication (pod test subsystem). The detection indication for the F_{oc} frequency shall be identical to that for the F_{ov}

frequency (see 3.5.1.2.2.4).

3.5.1.2.4 Self test subsystem. The electronics assembly shall contain s test capability which tests the sweep circuit, the detector circuit and battery condition as specified in 3.5.1.2.4.1 through 3.5.1.2.4.3.

3.5.1.2.4.1 Sweep circuit and detector circuit. When S2 is depressed, t sweep voltage shall be generated and applied to the input of the detector circuit, excluding the detector diode. The sweep voltage shall activate detector circuit and cause indicator lamp DS-1 to flash with a period of $(^{+1}, -.5)$ second and duty cycle of 10 to 25 percent during a portion of t active sweep. The number of flashes per sweep period shall be a function the adjustment of R6, R36, the battery voltage and sweep duration. The n of flashes per sweep shall be 10 ± 2 nominally, but may vary between 6 and

3.5.1.2.4.2 Battery test. When S1 or S2 is depressed, the DS-2 indicato lamp shall flash with a period of 2 ± 1 seconds and duty cycle of 10 to 2 percent when the battery voltage is 11.55 volts or above. It shall not f when the battery voltage is 11.50 volts or less.

3.5.1.2.4.3 Battery condition indicator threshold adjustment. Adjustment variable resistor, **R1**Z, shall provide a battery condition indicator thres range of at least 10.25 to 11.55 volts.

3.5.1.3 Plain shroud. The plain shroud shall be easily attachable/detac to/from the electronics assembly. It shall be manufactured in accordance Drawing 887AS400 and all documents listed thereon and shall meet all the requirements of that drawing. The shroud shall provide protection for the detector subsystem in the electronics assembly from high level electromagn fields and shall absorb and attenuate RF energy radiated from the antenna the unit under test. The shroud shall provide not less than 4 dB attenuat of the RF signal which is radiated from the weapon during a weapon test.

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3.5.1.4 EMCON shrouds. The EMCON shrouds shall be easily attachable /detachable to/from the electronics assembly. They shall be manufactured i accordance with Drawing 887AS500 or 1098AS5OO and all documents listed thereon and shall meet all the requirements of these drawings. The shrouds shall provide at least 48 dB attenuation of the RF signal radiated from the weapon during a weapon test. The EMCON shrouds shall be similar to the plain shroud except they shall have the features specified in 3.5.1.4.1 through 3.5.1.4.4.

3.5.1.4.1 Door and door operating mechanism. The 887AS500 EMCON shroud shall have a hinged door at the forward end and an opening - closing - locking mechanism suitable for opening or closing the door and locking it in both the closed and open positions.

3.5.1.4.2 RFI finger stock. The shrouds shall have fingerstock which shall contact a mating surface on the antenna of the system under test; the fingerstock shall be arranged for 360° contact with this surface.

3.5.1.4.3 Strap assembly. The EMCON shrouds shall have a strap and handle assembly for ho lding the shroud in place on the unit under test and applying pressure to the shroud/antenna interface to effect an EMI seal.

3.5.1.4.4 Electrical performance. When the shroud is attached to the electronics assembly and strapped onto the unit under test, the shroud shall reduce radiation from the test antenna by not less than 48 dB at any (and all) point(s) in space.

3.5.2 Weight. Heights shall be not greater than as listed in table 1:

TABLE I. Test set weights

Component	Drawing Number	Weight
Electronics assembly	1098AS1300	9 lb.
Plain shroud	887AS400	2.8 lb.
EMCON shroud	1098AS500	12 lb.
EMCON shroud	887AS500	4 lb.
Transit case	1098AS200	25 lb.
Battery pack	178AS200	17oz.

3.5.3 Content. The test set shall contain the major assemblies as specified in 3.5.3.1 through 3.5.3.4.

3.5.3.1 Electronics assembly (1098AS13OO). The electronics assembly shall contain the subassemblies specified in 3.5.3.1.1.

3.5.3.1.1 Chassis assembly (1098AS1302). The chassis assembly shall contain the printed circuit boards, subcarrier oscillators, voltage controlled oscillators, RF detector, battery pack, control switches, indicator lamps and associated cables.

3.5.3.1.2 Case assembly (1098AS34O). The case assembly shall contain the main case for the electronics assembly, the radiate and receive antennas, RF cables, radome, RF absorber material, and spare battery pack.

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3.5.3.1.3 Plain shroud assembly (887AS400). The plain shroud assembly shall contain a housing, radome, RF absorber and antenna hood.

3.5.3.3 EMCON shroud assemblies (887AS500 and 1098AS5OO). The EMCON shroud assembly shall contain a housing, radome, RF absorber, antenna hood, retainer strap assembly, and RFI seal subassembly which shall consist of RFI fingerstock and retainer, a hinged door and a door operating mechanism. EMCON shroud 1098AS500 has a removable cover, which can be used in place of the test set to allow use of shroud without test set, instead of a hinged door and a door operating mechanism.

3.5.3.4 Transit case (1098AS12OO). The transit case shall consist of the case assembly and cushioning material.

3.5.4 Controls. Two pushbutton control switches shall be located on the front panel. the “WPN TEST” pushbutton, when depressed shall enable the command, battery self test, and RF detection subsystems. The “SELF TEST” pushbutton, when depressed, shall enable the command and battery self test subsystems and shall apply a sweep voltage to activate the RF detector circuit. When both pushbuttons are depressed, the battery self test subsystem shall be enabled and the RF detection sensitivity shall be increased to provide an AN/AWW-7B pod radiation test capability. A six position rotary switch as specified in 3.4.1.5 shall be used to select either Phase I or Phase II function and a two digit pushbutton switch marked “CODE” shall be used to set the recognition code when in the Phase II mode.

3.5.5 Material. Materials shall be in accordance with DL 1098AS1OOO.

3.5.6 Workmanship Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All Items must meet all requirements of sections 3 and 5. he inspection set forth in this specification shall become a part of the contractor’ s’ overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The Inspection requiremtns specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.4.4).

4.3 Inspection conditions. Unless otherwise specifed herein, all inspections shall be performed in accordance with the test conditions specified in 3.3.4 and 3.3.5.

4.4 First article inspection. First article tests shall be made on a test set representative of the production test sets to be supplied under the contract. First article test performance shall be the responsibility of the contractor and shall be performed in accordance with approved test procedures (see 6.2.2) No first article inspections shall be conducted prior to acceptance of the first article test procedure by the contracting officer.

First article inspection performance requirements are those stated in 3.5.1.2 with environmental requirement stated in 4.9 and 3.3.5.

4.4.1 Scope of first article tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the test set meets all the requirements of this specification and the contract. These shall include environmental and interference tests in accordance with MIL-T-5422 as specified in MIL-T-28800. Test methods shall be in accordance with MIL-STD-81O except as modified by this specification.

4.4.2 First article approval. Approval of the first article will be by the procuring activity upon satisfactory completion of all tests (see 6.2.2).

4.4.3 Production equipments. Test sets supplied under the contract shall be, in all respects including design, construction, workmanship, performance and quality, equal to the approved first article sample. Each test set shall be capable of successfully completing the same tests as imposed on the first article sample.

4.4.4 Quality conformance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests specified below. All inspection and testing shall be to the approved quality conformance test procedures under the supervision of the Government Inspector

a. Individual tests (see 4.4.1)

b. Sampling tests (see 4.4.2) or reliability assurance tests (see 4.5.3)

c. Special tests (see 4.5)

d. Life tests (see 4.7)

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4.4.5 Individual tests. Each test set submitted for acceptance sh
subjected to the individual tests. These tests shall be adequate t
compliance with the requirements of material, workmanship, operatio
adequacy and reliability. As a minimum, each test set accepted sha
passed the following tests:

a. Examination of product (see 4.4.1.1)

b. Operational test (see 4.4.1.2)

c. Manufacturing run-in test (see 4.4.1.3)

4.4.5.1 Examination of product. Each test set shall be examined c
determine that the material and workmanship requirements have been

4.4.5.2 Operational test. Each test set shall be operated long en
permit the temperature to stabilize and to check characteristics an
data to assure required test set operation.

4,4.5.3 Manufacturing run-in test. Each test set shall be operate
conditions specified below for a period of 6 hours without failure.
shall be considered as anything which causes malfunctioning of the
Only those adjustments will be permitted which can be made using co
are accessible to the operator during normal use of the test set.

a.	Temperature	Standard Conditions
t).	Humidity	Standard Conditions
c.	Vibration	Any selected frequency withi of 20 to 30 Hz (excluding re points) and a minimum amplit Gs.

The test set shall be vibrated (without vibration isolators) for a
10 minutes prior to the beginning of the 6 hour period of operatio
feasible, the test set shall be operated during this vibration peri
purpose of detecting flaws and imperfect workmanship. Operation wi
specified limits of required performance is not necessarily require
the vibration period. The direction of vibration shall be vertical
normal mounting plane for 5 minutes and lateral to the plane for 5
Where it is not feasible to vibrate the test set in two directions,
vertical shall be used. During the 6 hour period of operation foll
10-minute vibration period, the test set shall be mechanically cycl
periodically through its various phases of operation. Should a fail
the test set shall be repaired and the test started over, except tha
minute vibration period need not be repeated when it is established
failure was not a result of vibration. Should failures occur repea

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corrective action shall be taken to eliminate this defect from future test sets. The 6 hour period specified above may be made up of several shorter periods to conform to normal working hours (see 6.2.2).

4.4.5.4 Temperature tests. Temperature tests shall be performed on the number of test sets specified in table II of electronics assembly (1098ASI3OO) at -40 and +55° C following the manufacturing run-in tests as follows:

a. Soak periods shall be 1 hour at -40 or at +55° C as applicable with test set unenergized.

b. A dc power supply capable of supplying 200 MA, 9.9 to 14.5 volts with 1 mv maximum ripple shall be substituted for the battery pack.

c. The battery test subsystem requirements (see 3.5.1.2.4.2) shall be verified with supply voltage at 11.25 ± .05 volts and at 11.60 ± .05 Volts .

d. The command subsystem requirements (see 3.5.1.2.1) and detector subsystem requirements (see 3.5.1.2.2 and 3.5.1.2.3) shall be verified with supply voltage at 11.60 ± .05 and at 14.25 ± .05 volts.

e. The self test subsystem (see 3.5.1.2.4) shall be verified at 11.60 ± .05 volts.

f. Following satisfactory conclusion of the temperature tests, the operational test shall again be performed at room ambient conditions using battery pack 178AS200.

TABLE II. Sampling test sample selection

Quantity of test sets offered for acceptance	Quantity to be selected for sampling tests
First 10	1
Next 50	1
Next 75	1
Next 100	One for each additional 200 or fraction thereof

4.4.6 Scope of tests. As a minimum, each test set selected for sampling tests shall be subjected to the following:

a. Complete operations test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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b. Operational test at certain environmental conditions. The conditions may vary for each test set tested and should be based on results of the first article, individual, and special tests.

c. Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 100 hours with no restriction on the number of failures (see 6.2.2).

4.4.7 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with the requirements of MIL-STD-781. Tests shall be conducted as required for both the qualification phase and the production acceptance (sampling) phase.

4.4.7.1 Reliability qualification phase. Prior to acceptance of test sets under the contract or purchase order, a contractually specified quantity of test sets (or a not less than three if not specified in 6.2.1) shall be subjected to the combined reliability qualification requirements for unsheltered shipboard equipment (category 3.B) of MIL-STD-781, using Test Plan IC therein θ_1 =200.

4.4.7.2 Reliability acceptance phase. Production lot samples of the test sets shall be subjected to combined production reliability acceptance requirements for unsheltered shipboard equipment (category 3.B) of MIL-STD-781, using Test Plan VC therein θ_1 =100).

4.4.7.2.1 Acceptance phase sample size. The entire contract quantity (except for those test sets used in the qualification phase) shall be considered one lot. The reliability acceptance sample size shall be 10 percent of the contract quantity, but not less than three units. The test program shall start the first month after the qualification phase has been completed. The samples shall be tested until an accept/reject decision has been reached. In the meantime, other test sets manufactured may be shipped in place, if acceptable to the procuring activity (see 6.2.2). The procuring activity reserves the right to stop the acceptance of test sets anytime after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the test set, parts, and quality control in order that the entire contract quantity will result in an accept decision.

4.4.7.3 Test details. The test details shall be part of the test procedures prior to the beginning of the qualification test phase of the reliability assurance tests (see 6.2.2).

4.4.7.3.1 Temperature extremes. The temperature extremes shall be limited to -40°c to +55°C storage, and -40°C to +55°C operating with batteries installed. The temperature extremes without batteries shall be -62° to +85°C (storage only).

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4.4.7.3.2 Voltage cycling. Voltage cycling in accordance with MIL-STD-781 is not required.

4.8 Special tests. Special tests shall be conducted on a quantity of test sets as specified by the Government inspector for the purpose of checking the effect of any design or material change on the performance of the test set and to assure quality control. The test sets selected for special tests may be from test sets previously subjected to the sampling or reliability assurance tests.

4.8.1 Special test schedule. Selection of test sets for special tests shall be made as follow:

a. On early test sets after an engineering or material change.

b. Whenever failure reports or other evidences indicate that additional tests are required. (This will be determined by the procuring activity.)

4.8.2 Scope of tests. Special tests shall consist of such tests acceptable to the procuring activity. Test procedures previously approved for the first article test may be used where applicable (see 6.2.2).

4.9 Test set failure. Should a failure occur during either the sampling reliability assurance or special tests, the following action shall be taken (see 6.2.2):

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or due to design defect.

c. Where practical, include a test (in the individual test) to check all test sets for this requirement until assurance is forthcoming that the defect has been corrected.

4.10 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on test sets that have passed the individual tests. The life test shall be performed under the conditions specified in 4.10.1. The life test sample will be selected by the Government inspector in accordance with table III. (Test sets which have successfully completed the sampling tests, reliability tests, or special tests may be selected for life tests.)

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TABLE III. Life test sample selection.

Quantity of test sets offered for acceptance	Quantity to be selected for life test
First 25	1
Next 175	1
Next 300	1 for each additional 500 or fraction thereof

When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of test sets equal to or more than that listed in table III receives at least 300 hours each of test time.

4.10.1 Life test conditions, The life test shall be conducted under the following simulated service conditions:

a.	Temperature	Room ambient (30°C ± 10°C)
b.	Altitude	Normal ground (O - 10,000 ft.)
c.	Humidity	Room ambient up to 90 percent relative humidity
d.	Input voltage	Standard Conditions
e.	Vibration	None

4.10.2 Life test period. The life test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the test set-temperature to stabilize. Periodically, the test set shall be turned on and off several times, and put through its various phases of operation.

4.10.3 Performance check. At approximately 8 hour intervals during the test, a limited performance check shall be made (see 6.2.2).

4.10.4 Test data. In the event of part failures, the defective part shall be replaced and the operation resumed for the remainder of the test period (see 6.2.2).

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4.10.4.1 Failure report. In the event of a failure, the Government Inspector shall be notified immediately (see 6.2.2).

4.11 Presubmission testing. No item, part, or complete test set shall be made available by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

4.12 Environmental tests. The test set shall operate as required when subjected to the following tests (see 3.3.2.1).

4.12.1 Temperature - altitude. The test set shall be tested in accordance with the following. Adjustment of controls shall be allowed. These conditions apply with batteries installed and only when not in conflict with 3.3.2.1.1 and 3.3.2.1.2.

a. With the test item de-energized, adjust the chamber test conditions to -40°C at 50,000 ft. After the test item temperature has stabilized, maintain these test conditions for at least 2 hours. Where it is possible, without changing the test conditions, the test item shall be visually inspected for the presence of deterioration which would impair future operation.

b. With the test item de-energized, adjust the chamber test conditions to an altitude at standard ambient. These chamber test conditions shall be maintained throughout this step. The test item shall be operated at 11.55 Vdc. The test item shall operate satisfactorily immediately following the specified warm-up time (see 3.3.3). The above operational sequence shall be repeated two additional times.

c. With the test item de-energized, adjust the chamber temperature to -40°C and altitude of 10,000 ft. The test item temperature shall be stabilized. The test item shall be energized at 14.60 Vdc and the chamber pressure adjusted to simulate 10,000 ft altitude. Upon reaching the specified chamber pressure and while maintaining the chamber temperature, the test item shall be checked for satisfactory operation (see 3.5.1.2) and the results recorded (see 6.2.2).

d. Adjust the chamber test conditions to standard ambient conditions at 55°C temperature. After the test item temperature has stabilized, an operational and performance check of the test item shall be made and shall meet the requirements of 3.5.1.2.

e. With the test item de-energized, adjust the chamber test conditions to 85°C and standard ambient altitude. The test item temperature shall be stabilized and then maintained for 16 hours. At the conclusion of this time period, where practicable, the test item shall be visually inspected for any deterioration.

f. Adjust the chamber test conditions to standard ambient conditions. After the test item temperature has stabilized, an operational and performance check of the test item shall be made and the results shall meet the requirements of 3.5.1.1.

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4.12.1.1 Summary. The following details shall be as specified in the equipment specifications:

a. Pretest data required.

b. Pertinent equipment operation parameters limits (input voltage).

c. Failure criteria.

4.12.2 Drip-proof. The test set shall be drip-proof tested in accordance with MIL-STD-108, except the test set shall be inclined only 15 degrees. The test set shall be operated during and after exposure.

4.12.3 Humidity. The test set shall be humidity tested in accordance with MIL-STD-810 Method 507.2, Procedure 1, except the maximum temperature shall be 55°C.

4.12.4 Salt spray. The test set shall be salt spray tested in accordance with MIL-STD-810, Method 509.2, Procedure 1.

4.12.5 Explosive-atmosphere. The test set shall be explosive tested in accordance with MIL-STD-810, Method 511.2, Procedure 1, except the maximum temperature and altitude shall be 55°C and 10,000 feet, respectively.

4.12.6 Vibration. The test set shall be subjected to vibration tests in accordance with MIL-T-28800 for Class 2, Style C equipment.

4.12.7 Shock. The test set shall be shock tested in accordance with MIL-STD-810, Method 516.1, Procedure 1, except the test set shall not be operated during the test.

4.13 Methods of examination and test. The procuring activity or Government inspector reserves the right to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract (see 6.2.2).

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5. PACKAGING

5.1 General. Packaging shall be in accordance with the general requirements of MIL-E-17555, supplemented as specified in 5.2 through 5.3.1.

5.2 Preservation-packaging. Unless otherwise specified in the contract or purchase order, preservation-packaging shall be in accordance with MIL-P-116, Method III (see 6.2.1).

5.3 Packing. Unless otherwise specified in the contract or purchase order, packing shall be level A (see 6.2.1). The packaged test sets shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, normal requirements. Sealing (taping) and reinforcing (banding) shall be in accordance with the appendix to PPP-B-636.

5.3.1 Unitized loads. Unitized loads, commensurate with the level of packing specified herein or in the contract, shall be used whenever the total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent possible. Exterior of the container shall be marked as follows:

THIS TEST SET CONTAINS NICKEL-CADMIUM BATTERIES
DO NOT STORE BELOW -40°F OR ABOVE +131°F
UNLESS BATTERIES ARE REMOVED

5.4 Marking. In addition to any special marking specified in the contract _ (see 6.2.1); each unit package, intermediate and exterior container, and unitized load shall be marked in accordance with MIL-STD-129. When specified (see 6.2.1), the marking of domestic shipments for civil agencies shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The intended use of the Guided Weapon Test Set AN/DSM-139C is to perform a GO/NO-GO confidence check on the data link subsystem in data link modified Walleye weapons after the weapon has been installed on the carriage aircraft. The test set exercises both the command and video portions of the weapon data link subsystem. The test set also has the capability to detect RF radiation from the AN/AWW-7B, AN/AWW-9, or AN/AWW-12 pods.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. F_{OC} frequency (see 3.5.1.2.1)

c. F_{OV} frequency (see 3.5.1.2.2.1)

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d. Selection of applicable levels of packaging, packing, and marking (see 5.3, 5.3.1, 5.4)

e. When a first article is required for inspection and approval (see 4.4)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID no.	Optior
4.4, 4.4.2	Report, First Article Inspection	DI-T-4902	---
4.10.4.1	Failure Report	UDI-T-23893	---
4.4.5.3	Report, Burn-in Test	DI-R-7040	---
4.4.7.2.1. 4.4.7.3	Procedures, Reliability Tests	DI-R-7035	---
4.8.2, 4.13, 4.10.3	Procedures, Production/ Acceptance Inspection	DI-T-4903	---
4.9	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719	---
4.13, 4.10.4	Test Report	DI-T-5247	---

(Data Item descriptions related to this specification, and identified In section will be approved and listed as such in DoD 5010.12-L AMSDL. Copies of item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required the item will be tested and should be a first article sample (or it may be a standard production from the contractor’s current inventory as specified In 4.1.1). The first artic should consist of one test set unless otherwise specified in the contract or purchase order. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids sh provide that the Government reserves the right to waive the requirement for samp for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such

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products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval Is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Antenna
Antenna hood
Battery pack
Code switches
Command receivers
Control-monitor set
Data link
Guided weapon
Phase I
Phase II
Pod
Radome
Receive antenna
RF carrier
RF signal
Shroud assembly
Sweep circuit
Test set
Test Set, Guided Weapon AN/DSM-139C
Transit case
Video signal
Voltage controlled oscillator
Walleye weapon
Weapon recognition code
WPN TEST switch

Preparing Activity:
Navy - AS
(Project 4935-N058

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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5. PROBLEM AREAS

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